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Wireless dongle

Abstract

A wireless dongle includes a circuit board, a connector, a wireless module and a printed antenna. The circuit board has a first front edge, a first rear edge, a first left edge and a first right edge. The connector is disposed to a middle of a front of the circuit board. The wireless module is disposed to a middle area and a left area of the circuit board. The connector is electrically connected to the wireless module. The printed antenna is positioned at a right area of the circuit board. The printed antenna is a monopole antenna. The wireless module is connected between the connector and the printed antenna. The printed antenna has a feed-in section positioned close to the first rear edge of the circuit board, and a radiation section.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS
(1) The present application is based on, and claims priority from, China Patent Application No. 202321160391.4, filed May 15, 2023, the disclosure of which is hereby incorporated by reference herein in its entirety.
BACKGROUND OF THE INVENTION
Field of the Invention
(2) The present invention generally relates to a wireless dongle, and more particularly to a wireless dongle having a printed antenna with a compact layout.
Description of Related Art
(3) With the rapid development of mobile communication in recent years, various mobile devices gradually tend to be wireless. Simultaneously, people expect the mobile devices and peripheral devices to be thinner, lighter and smaller, so that the mobile devices and peripheral devices are easy to be carried. The mobile devices and the peripheral devices are prompted to develop towards miniaturized sizes, in that case, the sizes of the mobile devices and the peripheral devices are smaller and smaller, for example, a size of a wireless adapter is getting smaller and smaller, and an antenna which is disposed inside the wireless adapter must be correspondingly reduced, so a demand for the antenna that is able to work stably in provided frequency bands and has the smaller size is increased.
(4) Therefore, it is necessary to provide a wireless dongle having a printed antenna with a compact layout, the wireless dongle works in a limited space, and the wireless dongle is able to stably work in a predetermined frequency band.
BRIEF SUMMARY OF THE INVENTION
(5) An object of the present invention is to provide a wireless dongle having a compact layout of a printed antenna. The wireless dongle includes a circuit board, a connector, a wireless module and a

printed antenna. The circuit board has a first front edge, a first rear edge opposite to the first front edge, a first left edge, and a first right edge opposite to the first left edge. The connector is disposed to a middle of a front of the circuit board, and the connector projects beyond a middle of the first front edge of the circuit board. The wireless module is disposed to a middle area and a left area of the circuit board. The connector is electrically connected to the wireless module. The printed antenna is positioned at a right area of the circuit board. The printed antenna is a monopole antenna. The wireless module is connected between the connector and the printed antenna. The wireless module is disposed between the printed antenna and the first left edge of the circuit board. The printed antenna is disposed between the wireless module, and the first right edge of the circuit board. The printed antenna is arranged away from the connector. The printed antenna has a feed-in section positioned close to the first rear edge of the circuit board, and a radiation section. The feed-in section is connected to a right rear corner of the wireless module. The radiation section is horizontally extended rightward from a right end of the feed-in section along the first rear edge of the circuit board, then is bent frontward and is extended towards the first front edge of the circuit board, later is bent rightward and is horizontally extended towards the first right edge of the circuit board and is further bent rearward and is longitudinally extended close to the first rear edge of the circuit board. A second rear edge of the radiation section is separated from the first rear edge of the circuit board by a first clearance, a second front edge of the radiation section is separated from the first front edge of the circuit board by a second clearance, an outer edge of the radiation section is separated from the first right edge of the circuit board by the first clearance, the second clearance is greater than the first clearance, an extending path of a front portion of the radiation section forms in an inverted U shape, a tail end of the radiation section is located at a right rear corner of the circuit board.

(6) Another object of the present invention is to provide a wireless dongle. The wireless dongle includes a circuit board, a connector, a wireless module and a printed antenna. The circuit board has a first front edge, a first rear edge opposite to the first front edge, a first left edge, and a first right edge opposite to the first left edge. The connector is disposed to a middle of a front of the circuit board, and a front of the connector projects beyond a middle of the first front edge of the circuit board. The wireless module is disposed to a middle area and a left area of the circuit board. A rear of the connector is disposed to a front of the wireless module. The wireless module contains a plurality of circuits. The plurality of the circuits include a first wireless unit and a plurality of second wireless units. The first wireless unit is disposed to a right front corner of the wireless module. The first wireless unit and the plurality of the second wireless units surround the rear of the connector. The connector is electrically connected to the wireless module. The printed antenna is positioned at a right area of the circuit board. The printed antenna is a monopole antenna. The wireless module is connected between the connector and the printed antenna. The printed antenna has a feed-in section positioned close to the first rear edge of the circuit board, and a radiation section. The feed-in section is connected to a right rear corner of the wireless module. The radiation section is horizontally extended rightward from a right end of the feed-in section, then is bent frontward and is extended towards the first front edge of the circuit board, later is bent rightward and is horizontally extended towards the first right edge of the circuit board and is further bent rearward and is longitudinally extended close to the first rear edge of the circuit board. A second rear edge of the radiation section is separated from the first rear edge of the circuit board by a first clearance, a second front edge of the radiation section is separated from the first front edge of the circuit board by a second clearance, an outer edge of the radiation section is separated from the first right edge of the circuit board by the first clearance, the second clearance is greater than the first clearance, a front portion of the radiation section is inverted U-shaped, two facing surfaces of the inverted U-shaped front portion of the radiation section are separated by a third clearance, a tail end of the radiation section is located at a right rear corner of the circuit board.

(7) Another object of the present invention is to provide a wireless dongle. The wireless dongle

includes a circuit board, a connector, a wireless module and a printed antenna. The circuit board has a first front edge, a first rear edge opposite to the first front edge, a first left edge, and a first right edge opposite to the first left edge. The connector is disposed to a middle of a front of the circuit board, and the connector projects beyond a middle of the first front edge of the circuit board. The wireless module is disposed to a middle area and a left area of the circuit board. A rear of the connector is disposed to a front of the wireless module. The connector is electrically connected to the wireless module. The wireless module contains a plurality of circuits. The plurality of the circuits surround the rear of the connector. The printed antenna is positioned at a right area of the circuit board. The printed antenna is a monopole antenna. The wireless module is connected between the connector and the printed antenna. The plurality of the circuits are disposed between the connector and the printed antenna. The wireless module is disposed between the printed antenna, and the first left edge of the circuit board. The printed antenna is disposed between the wireless module, and the first right edge of the circuit board. The printed antenna is arranged away from the connector. The printed antenna has a feed-in section positioned close to the first rear edge of the circuit board, and a radiation section. The feed-in section is connected to a right rear corner of the wireless module. The plurality of the circuits are located to one side of the feed-in section. The radiation section is horizontally extended rightward from a right end of the feed-in section along the first rear edge of the circuit board, then is bent frontward and is extended towards the first front edge of the circuit board, later is bent rightward and is horizontally extended towards the first right edge of the circuit board and is further bent rearward and is longitudinally extended close to the first rear edge of the circuit board. A second rear edge of the radiation section is separated from the first rear edge of the circuit board by a first clearance, a second front edge of the radiation section is separated from the first front edge of the circuit board by a second clearance, an outer edge of the radiation section is separated from the first right edge of the circuit board by the first clearance, the second clearance is greater than the first clearance, an extending path of a front portion of the radiation section forms in an inverted U shape, a tail end of the radiation section is located at a right rear corner of the circuit board.
(8) As described above, the wireless dongle has a compact layout of the printed antenna, so the printed antenna of the wireless dongle works in a limited space, and the printed antenna of the wireless dongle is able to stably work in a predetermined frequency band. As a result, the wireless dongle is adapted to a miniaturization development trend of electronic products, and a wireless development trend of the electronic products.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present invention will be apparent to those skilled in the art by reading the following description, with reference to the attached drawings, in which:
(2) FIG. **1** is a front diagram of a wireless dongle according to a preferred embodiment of the present invention;
(3) FIG. **2** is a voltage standing wave ratio (VSWR) chart of a printed antenna of the wireless dongle according to the preferred embodiment of the present invention;
(4) FIG. **3** is a smith chart of the printed antenna of the wireless dongle according to the preferred embodiment of the present invention; and
(5) FIG. **4** is a return loss chart of the printed antenna of the wireless dongle according to the preferred embodiment of the present invention.
DETAILED DESCRIPTION OF THE INVENTION
(6) Referring to FIG. **1**, a wireless dongle **100** in accordance with a preferred embodiment of the present invention is shown. The wireless dongle **100** includes a circuit board **10**, a connector **20**, a

wireless module **30** and a printed antenna **40**. The connector **20**, the wireless module **30** and the printed antenna **40** are all arranged on the circuit board **10**. The connector **20** is electrically connected to the wireless module **30**, and the wireless module **30** is electrically connected to the printed antenna **40**. The wireless module **30** is connected between the connector **20** and the printed antenna **40**.

(7) The circuit board **10** is formed in a rectangular shape. The circuit board **10** has two opposite long edges and two opposite short edges. In this preferred embodiment, the circuit board **10** has a first front edge **101**, a first rear edge **102** opposite to the first front edge **101**, a first left edge **103**, and a first right edge **104** opposite to the first left edge **103**. The first left edge **103** is connected between two left ends of the first front edge **101** and the first rear edge **102**, and the first right edge **104** connected between two right ends of the first front edge **101** and the first rear edge **102**. The first front edge **101** and the first rear edge **102** of the circuit board **10** are the two opposite long edges. The first left edge **103** and the first right edge **104** of the circuit board **10** are the two opposite short edges.

(8) The connector **20** is disposed to a middle of a front of the circuit board **10**, and a front of the connector **20** projects beyond a middle of the first front edge **101** of the circuit board **10**. The wireless module **30** is disposed to a middle area and a left area of the circuit board **10**. The printed antenna **40** is positioned at a right area of the circuit board **10**. The wireless module **30** is disposed between the printed antenna **40** and the first left edge **103** of the circuit board **10**. Specifically, the wireless module **30** is disposed between a middle of the printed antenna **40**, and the first left edge **103** of the circuit board **10**. The printed antenna **40** is disposed between the wireless module **30**, and the first right edge **104** of the circuit board **10**. The printed antenna **40** is arranged away from the connector **20**.

(9) After a user inserts the wireless dongle **100** into a mobile device (not shown), the mobile device is able to proceed with wireless communication via the wireless dongle **100**. The mobile device is a notebook or a smart phone. In this preferred embodiment, a distance between the mobile device and the printed antenna **40**, and a distance between the mobile device and a metal shell of the connector **20** all affect characteristics of the printed antenna **40**, such as a voltage standing wave ratio, and a field pattern and reflection loss, so under a premise that a layout of the printed antenna **40** is cooperated with a circuit layout of the wireless module **30**, the layout of the printed antenna **40** need be far away from the metal shell of the connector **20** and the mobile device.

(10) The wireless module **30** is able to be a Bluetooth module or a 2.4G module. The wireless module **30** contains a plurality of circuits **31** to proceed with data transmission and data conversion, and the plurality of the circuits **31** are selected from a group consisting of micro-controllers, filters and sensors. The plurality of the circuits **31** are able to be the micro-controllers, the filters, the sensors, different type circuits, or different type components, etc. A quantity of the circuits **31** shown in FIG. **1** are just schematic, and the quantity of the circuits **31** should be without being limited to the quantity of the circuits **31** which is shown in FIG. **1**. The plurality of the circuits **31** include a first wireless unit **301** and a plurality of second wireless units **302**. The first wireless unit **301** is disposed to a right front corner of the wireless module **30**. A rear of the connector **20** is disposed to a front of the wireless module **30**. In this preferred embodiment, the plurality of the circuits **31** are disposed between the connector **20** and the printed antenna **40**. The plurality of the circuits **31** surround the rear of the connector **20**. The first wireless unit **301** and the plurality of the second wireless units **302** surround the rear of the connector **20**.

(11) The printed antenna **40** is a monopole antenna. The printed antenna **40** has a feed-in section **41** and a radiation section **42**. The feed-in section **41** is positioned close to the first rear edge **102** of the circuit board **10**. The feed-in section **41** is connected to a right rear corner of the wireless module **30**. The radiation section **42** is horizontally extended rightward from a right end of the feed-in section **41** along the first rear edge **102** of the circuit board **10**, then is bent frontward and is extended towards the first front edge **101** of the circuit board **10**, later is bent rightward and is

horizontally extended towards the first right edge **104** of the circuit board **10** and is further bent rearward and is longitudinally extended close to the first rear edge **102** of the circuit board **10**. The plurality of the circuits **31** are located to one side of the feed-in section **41**.

(12) A second rear edge **401** of the radiation section **42** is separated from the first rear edge **102** of the circuit board **10** by a first clearance s**1**. A second front edge **402** of the radiation section **42** is separated from the first front edge **101** of the circuit board **10** by a second clearance s**2**. An outer edge **403** of the radiation section **42** is separated from the first right edge **104** of the circuit board **10** by the first clearance s**1**. The second clearance s**2** is greater than the first clearance s**1**. An extending path of a front portion of the radiation section **42** forms in an inverted U shape. The front portion of the radiation section **42** is inverted U-shaped. Two facing surfaces of the inverted U-shaped front portion of the radiation section **42** are separated by a third clearance s**3**. A tail end of the radiation section **42** is located at a right rear corner of the circuit board **10**.

(13) Referring to FIG. **1** again, in this preferred embodiment, the radiation section **42** includes a first zone **42***a* straightly extended rightward from the right end of the feed-in section **41**, a second zone **42***b* straightly extended frontward from a tail end of the first zone **42***a*, a third zone **42***c* straightly extended rightward from a tail end of the second zone **42***b*, and a fourth zone **42***d* straightly extended rearward from a tail end of the third zone **42***c*. A second left edge **404** of the fourth zone **42***d* is separated from a second right edge **405** of the second zone **42***b* by the third clearance s**3**. A tail end of the fourth zone **42***d* is located at the right rear corner of the circuit board **10**.

(14) The second rear edge **401** of the first zone **42***a* of the radiation section **42** is separated from the first rear edge **102** of the circuit board **10** by the first clearance s**1**. The second front edge **402** of the third zone **42***c* of the radiation section **42** is separated from the first front edge **101** of the circuit board **10** by the second clearance s**2**. The outer edge **403** of the fourth zone **42***d* of the radiation section **42** is separated from the first right edge **104** of the circuit board **10** by the first clearance s**1**. The first zone **42***a* and the third zone **42***c* are extended horizontally, and the first zone **42***a* and the third zone **42***c* are formed in transverse strip shapes. An extending path of the first zone **42***a* is parallel to an extending path of the third zone **42***c*. The first zone **42***a* is paralleled to the third zone **42***c*. The first zone **42***a* and the third zone **42***c* are parallel to the first front edge **101** and the first rear edge **102** of the circuit board **10**. The second zone **42***b* and the fourth zone **42***d* are extended longitudinally, and the second zone **42***b* and the fourth zone **42***d* are formed in longitudinal strip shapes. An extending path of the second zone **42***b* is paralleled to an extending path of the fourth zone **42***d*. The second zone **42***b* is paralleled to the fourth zone **42***d*.

(15) According to the above-mentioned description configuration, the feed-in section **41** and the radiation section **42** of the printed antenna **40** are alternately transmitted, and electric fields and magnetic fields of the feed-in section **41** and the radiation section **42** are interacted to oscillate electromagnetic waves in a frequency band which is ranged from 2.4 GHz to 2.5 GHz. In practice, the first clearance s**1** is 0.2 mm, the second clearance s**2** is approximately equal to the distance between the mobile device and the printed antenna **40**, the second clearance s**2** is 0.5 mm, and the third clearance s**3** is 2 mm.

(16) When the printed antenna **40** of the wireless dongle **100** according to the present invention is used in wireless communication, a current from the wireless module **30** is fed by the feed-in section **41**. The current passes through the radiation section **42**, the electromagnetic waves in the frequency band which is ranged from 2.4 GHz to 2.5 GHz are oscillated. Thus, the printed antenna **40** of the wireless dongle **100** works in a limited space, and the printed antenna **40** of the wireless dongle **100** is able to stably work in a predetermined frequency band.

(17) Referring to FIG. **1** to FIG. **3**, a voltage standing wave ratio (VSWR) chart of the printed antenna **40** of the wireless dongle **100** according to the present invention is shown in FIG. **2**. A smith chart of the printed antenna **40** of the wireless dongle **100** according to the present invention is shown in FIG. **3**. When the printed antenna **40** is operated at 2.4 GHz, a VSWR value of the

printed antenna **40** is 3.2842 which is shown at a position M**1** of FIG. **2** and FIG. **3**. When the printed antenna **40** is operated at 2.45 GHZ, a VSWR value of the printed antenna **40** is 2.6379 which is shown at a position M**2** of FIG. **2** and FIG. **3**. When the printed antenna **40** is operated at 2.5 GHZ, a VSWR value of the printed antenna **40** is 2.2588 which is shown at a position M**3** of FIG. **2** and FIG. **3**. Therefore, the printed antenna **40** of the wireless dongle **100** according to the present invention is able to be stably operated in the frequency band which is ranged from 2.4 GHz to 2.5 GHZ.
(18) Referring to FIG. **1** and FIG. **4**, when the printed antenna **40** of the wireless dongle **100** is operated in the frequency band which is ranged from 2.4 GHz to 2.5 GHZ, a return loss of a bandwidth of the printed antenna **40** is roughly within −10 dB, a loss degree of the printed antenna **40** is small, and a radiation energy of the printed antenna **40** is large.
(19) As described above, the wireless dongle **100** has the printed antenna **40** with a compact layout, so the printed antenna **40** of the wireless dongle **100** works in the limited space, and the printed antenna **40** of the wireless dongle **100** is able to stably work in the predetermined frequency band. As a result, the wireless dongle **100** is adapted to a miniaturization development trend of electronic products, and a wireless development trend of the electronic products.
(20) Though the present invention is disclosed as the above-mentioned preferred embodiment, the preferred embodiment disclosed in this invention is without being intended to limit a scope of this invention. In related technical fields, anyone with ordinary knowledges should be able to make a few changes and embellishments within a spirit and a protection scope of this invention, so the protection scope of this invention should regard defined claims of an attached application patent as a standard.

Claims

1. A wireless dongle, comprising: a circuit board having a first front edge, a first rear edge opposite to the first front edge, a first left edge, and a first right edge opposite to the first left edge; a connector disposed on a middle of a front of the circuit board, the connector projecting beyond a middle of the first front edge of the circuit board; a wireless module disposed on a middle area and a left area of the circuit board, the connector being electrically connected to the wireless module; and a printed antenna positioned at a right area of the circuit board, the printed antenna being a monopole antenna, the wireless module being connected between the connector and the printed antenna, the wireless module being disposed between the printed antenna and the first left edge of the circuit board, the printed antenna being disposed between the wireless module and the first right edge of the circuit board, the printed antenna being arranged away from the connector, the printed antenna having: a feed-in section positioned close to the first rear edge of the circuit board, the feed-in section being connected to a right rear corner of the wireless module; and a radiation section horizontally extended rightward from a right end of the feed-in section along the first rear edge of the circuit board, then being bent frontward and extended towards the first front edge of the circuit board, later being bent rightward and horizontally extended towards the first right edge of the circuit board and being further bent rearward and longitudinally extended close to the first rear edge of the circuit board; wherein a second rear edge of the radiation section is separated from the first rear edge of the circuit board by a first clearance, a second front edge of the radiation section is separated from the first front edge of the circuit board by a second clearance, an outer edge of the radiation section is separated from the first right edge of the circuit board by the first clearance, the second clearance is greater than the first clearance, an extending path of a front portion of the radiation section forms in an inverted U shape, and a tail end of the radiation section is located at a right rear corner of the circuit board.
2. The wireless dongle as claimed in claim 1, wherein the radiation section includes a first zone straightly extended rightward from the right end of the feed-in section, a second zone straightly

extended frontward from a tail end of the first zone, a third zone straightly extended rightward from a tail end of the second zone, and a fourth zone straightly extended rearward from a tail end of the third zone, a second left edge of the fourth zone is separated from a second right edge of the second zone by a third clearance.

3. The wireless dongle as claimed in claim 2, wherein the second rear edge of the first zone of the radiation section is separated from the first rear edge of the circuit board by the first clearance, the second front edge of the third zone of the radiation section is separated from the first front edge of the circuit board by the second clearance, the outer edge of the fourth zone of the radiation section is separated from the first right edge of the circuit board by the first clearance.

4. The wireless dongle as claimed in claim 2, wherein a tail end of the fourth zone is located at the right rear corner of the circuit board.

5. The wireless dongle as claimed in claim 2, wherein the first zone and the third zone are extended horizontally, and the first zone and the third zone are formed in transverse strip shapes, the first zone is paralleled to the third zone, the first zone and the third zone are parallel to the first front edge and the first rear edge of the circuit board.

6. The wireless dongle as claimed in claim 2, wherein the second zone and the fourth zone are extended longitudinally, and the second zone and the fourth zone are formed in longitudinal strip shapes, the second zone is paralleled to the fourth zone.

7. The wireless dongle as claimed in claim 2, wherein an extending path of the first zone is parallel to an extending path of the third zone, an extending path of the second zone is paralleled to an extending path of the fourth zone.

8. A wireless dongle, comprising: a circuit board having a first front edge, a first rear edge opposite to the first front edge, a first left edge, and a first right edge opposite to the first left edge; a connector disposed on a middle of a front of the circuit board, a front of the connector projecting beyond a middle of the first front edge of the circuit board; a wireless module disposed on a middle area and a left area of the circuit board, a rear of the connector being disposed to a front of the wireless module, the wireless module containing a plurality of circuits, the plurality of the circuits including a first wireless unit and a plurality of second wireless units, the first wireless unit being disposed to a right front corner of the wireless module, the first wireless unit and the plurality of the second wireless units surrounding the rear of the connector, the connector being electrically connected to the wireless module; and a printed antenna positioned at a right area of the circuit board, the printed antenna being a monopole antenna, the wireless module being connected between the connector and the printed antenna, the printed antenna having: a feed-in section positioned close to the first rear edge of the circuit board, the feed-in section being connected to a right rear corner of the wireless module; and a radiation section horizontally extended rightward from a right end of the feed-in section, then being bent frontward and extended towards the first front edge of the circuit board, later being bent rightward and horizontally extended towards the first right edge of the circuit board and being further bent rearward and longitudinally extended close to the first rear edge of the circuit board; wherein a second rear edge of the radiation section is separated from the first rear edge of the circuit board by a first clearance, a second front edge of the radiation section is separated from the first front edge of the circuit board by a second clearance, an outer edge of the radiation section is separated from the first right edge of the circuit board by the first clearance, the second clearance is greater than the first clearance, a front portion of the radiation section is inverted U-shaped, two facing surfaces of the inverted U-shaped front portion of the radiation section are separated by a third clearance, and a tail end of the radiation section is located at a right rear corner of the circuit board.

9. A wireless dongle, comprising: a circuit board having a first front edge, a first rear edge opposite to the first front edge, a first left edge, and a first right edge opposite to the first left edge; a connector disposed on a middle of a front of the circuit board, the connector projecting beyond a middle of the first front edge of the circuit board; a wireless module disposed on a middle area and

a left area of the circuit board, a rear of the connector being disposed to a front of the wireless module, the connector being electrically connected to the wireless module, the wireless module containing a plurality of circuits, the plurality of the circuits surrounding the rear of the connector; and a printed antenna positioned at a right area of the circuit board, the printed antenna being a monopole antenna, the wireless module being connected between the connector and the printed antenna, the plurality of the circuits being disposed between the connector and the printed antenna, the wireless module being disposed between the printed antenna and the first left edge of the circuit board, the printed antenna being disposed between the wireless module and the first right edge of the circuit board, the printed antenna being arranged away from the connector, the printed antenna having: a feed-in section positioned close to the first rear edge of the circuit board, the feed-in section being connected to a right rear corner of the wireless module, the plurality of the circuits being located to one side of the feed-in section; and a radiation section horizontally extended rightward from a right end of the feed-in section along the first rear edge of the circuit board, then being bent frontward and extended towards the first front edge of the circuit board, later being bent rightward and horizontally extended towards the first right edge of the circuit board and being further bent rearward and longitudinally extended close to the first rear edge of the circuit board; wherein a second rear edge of the radiation section is separated from the first rear edge of the circuit board by a first clearance, a second front edge of the radiation section is separated from the first front edge of the circuit board by a second clearance, an outer edge of the radiation section is separated from the first right edge of the circuit board by the first clearance, the second clearance is greater than the first clearance, an extending path of a front portion of the radiation section forms in an inverted U shape, and a tail end of the radiation section is located at a right rear corner of the circuit board.